Name: ______________________________

Instructor: ______________________________

CHE 225 – Introduction to Chemical Engineering Analysis

Practice Test #2. Closed book – closed notes. Write all your answers in the provided paper. *Begin each numbered problem on a new page, show all your work, and box your final numerical answers*. Only write your answers on the front of each page and start each problem on a fresh page.

HONOR PLEDGE: "I have neither given nor received unauthorized aid on this test. I will not discuss the exam with anyone until after 5 pm on Friday, xxx, xxx."

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Your name (printed)

____

Problem 1 (38 points)

The virial equation of state (EOS) is frequently used to relate the gas pressure (P), specific molar volume ($\hat{V}$), and temperature (T) for non-ideal gases. The virial EOS is typically expressed as:

$$\frac{P\hat{V}}{RT} = 1 + \frac{B}{\hat{V}}$$

where B is a constant only dependent on temperature T.

For given P and T, we would like to calculate the specific molar volume ($\hat{V}$) by using the Newton-Raphson method. A step-by-step procedure is described below, and please answer questions for each step:

a) (**4 points**) First, the virial EOS needs to be rearranged into a new expression, called f($\hat{V}$). Please write the correct expression of f($\hat{V}$) on the answer sheet.

b) (**8 points**) Second, please write out the expression of the derivative of f($\hat{V}$), called f ’($\hat{V}$).

c) (**10 points**) Third, Newton-Raphson method is a numerical iteration approach, where the old $\hat{V}_i$ will be repeatedly replaced by new $\hat{V}_{i+1}$ to approach the correct solution. Please write the **iteration equation** between $\hat{V}_{i+1}$ and $\hat{V}_i$ that is used in the Newton-Raphson method. <u>Substitute the variables in the iteration equation by using the expressions derived in (a) and (b).</u>

d) (**10 points**) Finally, Newton-Raphson method will continue to iterate until a certain condition is met. Please both describe the condition in a single sentence **<u>and</u>** list the expression of the condition, by which Newton-Raphson iteration will stop.

e) (**3 points**) Name another numerical iteration method that can be used to solve nonlinear equations.

f) (**3 points**) What is the unit of B? Assume that the units of P is atm, T is K, $\hat{V}$ is L/mol, and R is L • atm/ (mol • K).

Problem 2 (16 points)

Dr. Genzer is planning to use Matlab to solve the following nonlinear equation:

$$\cos(x) = x^3 \ln(x)$$

Please answer the questions below to help Dr. Genzer walk through the process:

a) (**4 points**) In order to apply fzero or fsolve function in the Matlab, Dr. Genzer needs to rearrange the nonlinear equation at the first place. Please write down the rearranged function of the original nonlinear equation: y = _________________;

b) (**4 points**) Dr. Genzer then needs to define that function in the Matlab, and would like to give it a name of “nonlinearFunc”. He drafted the following script, but forgot where he should type in.

```
function y = nonlinearFunc(x)
y = _______________ ;
```

The above script should be typed into the ____________ (Editor or Command, select one) window.

c) (**4 points**) Finally, Dr. Genzer will run the following command to calculate the x:

```
fsolve(@nonlinearFunc,1)
```

What is the purpose to include a number, 1, in the command line? Please explain.

d) (**4 points**) Are there alternative ways to use the function *fsolve* other than the one described in c)? If so, please specify.

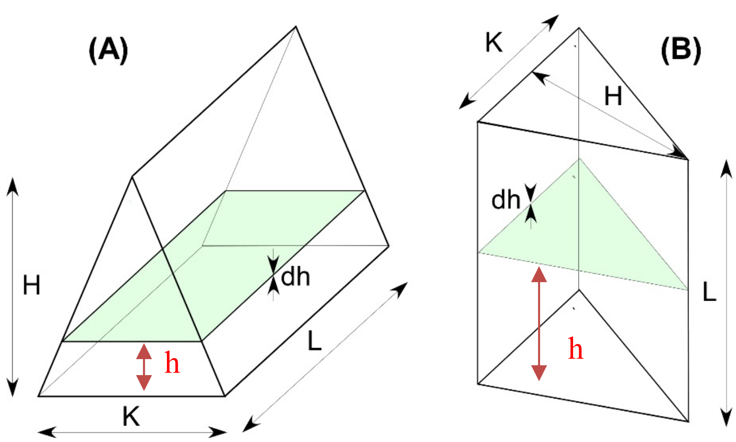
Problem 3 (**30 points**)

Consider a triangular prism as depicted in figures (A & B) below.

a) (**20 points**) Derive an expression for the overall volume of the object in figure A, assuming that you cut it into individual horizontal slices of thickness dh and add all slices together starting from h=0 all the way to h=H.

b) (**7 points**) If the triangular prism is standing vertically, as shown in figure (B), derive an expression for the volume of the object again. [Note: Although a thin slice is shown in (B), you do not need to use the same approach as before in your derivation. You can write down the answer directly].

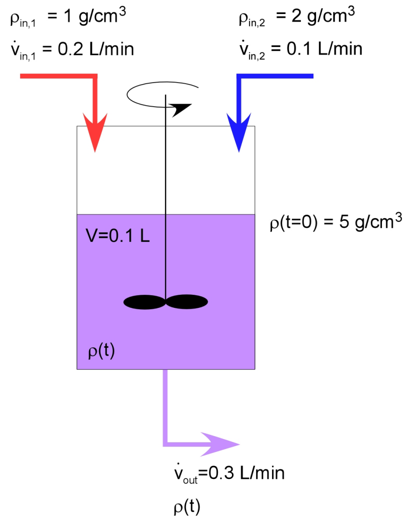
c) (**3 points**) Compare results obtained in (A) and (B).



Problem 4 (25 points)

A continuous stirred tank (CST) contains 0.1 L of a liquid having a density of 5 g/cm^3. At time t=0 seconds, the tank is being filled with two separate input streams, each comprising a different liquid. One stream has a density of 1 g/cm^3 and is being supplied with the volumetric flow of 0.2 L/min. The second stream contains a liquid with a density of 2 g/cm^3 and is being supplied with the volumetric flow of 0.1 L/min. The exit steam leaves the CST at a volumetric flow of 0.3 L/min.

The system set up is depicted graphically in the figure below.



Assuming that the volume of the liquid in the CTS remains constant, perform the following tasks:

a) (**10 points**) Starting from the overall mass balance for transient systems, derive a mass balance equation that describes the variation of density (ρ) of the liquid inside the CST as a function of time. Do not solve the ODE.

b) (**5 points**) Using the result in part a) determine the final density of the liquid inside the CST at *steady state*. Pay attention to the units.

c) (**10 points**) Solve the equation determined in part a) analytically to obtain an expression for the variation of liquid density (ρ) inside the CST as a function of time. Pay attention to the units.

Table 1. Unit Conversions & Constants

Quantity	Conversion
Energy	1 J = 1 Nm = 10^7 ergs = 10^7 dyne·cm = 2.778 x 10^{-7} kW·h
Force	1 N = 1 kg·m/s^2 = 10^5 dyne (or g*m/s^2)
Length	1 m = 100 cm = 39.37 in = 3.2808 ft
Mass	1 kg = 1000 g = 2.20462 lb_m
Power	1 W = 1 J/s = 0.23901 cal/s
Pressure	1 atm = 1.01325 x 10^5 Pa (or N/m^2) = 760 mm Hg = 14.696 lb_f/in^2 (or psi) = 1.01325 bars
Temperature	T(K) = T(ºC)+273.15; T(ºF) = 1.8·T(ºC)+32
Volume	1 m^3 = 1000 liters = 10^6 cm^3 = 35.3145 ft^3
Time	1 h = 60 min = 3600 s
Universal gas constant	8.314 J/K/mol 8.206 m^3*atm/K/mol 1.987 kcal/K/mol

Table 2. Derivative Rules

Function name	Function	Derivative
	$f(x)$	$f'(x)$
Constant	*const*	0
Linear	x	1
Power	x^a	$a x^{a-1}$
Exponential	e^x	e^x
Exponential	a^x	$a^x \ln a$
Natural logarithm	$\ln(x)$	$\frac{1}{x}$
Logarithm	$\log_b(x)$	$\frac{1}{x\, ln(b)}$
Sine	$\sin x$	$\cos x$
Cosine	$\cos x$	$-\sin x$
Tangent	$\tan x$	$\frac{1}{\cos^2(x)}$
Arcsine	$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$

Arccosine	arccos x	$-\frac{1}{\sqrt{1-x^2}}$
Arctangent	arctan x	$\frac{1}{1+x^2}$
Hyperbolic sine	sinh x	cosh x
Hyperbolic cosine	cosh x	sinh x
Hyperbolic tangent	tanh x	$\frac{1}{\cosh^2(x)}$
Inverse hyperbolic sine	$\sinh^{-1} x$	$\frac{1}{\sqrt{x^2+1}}$
Inverse hyperbolic cosine	$\cosh^{-1} x$	$\frac{1}{\sqrt{x^2-1}}$
Inverse hyperbolic tangent	$\tanh^{-1} x$	$\frac{1}{1-x^2}$

Table 3. Common integrals

Common Functions	Function	Integral
Constant	$\int a\, dx$	$ax + C$
Variable	$\int x\, dx$	$x^2/2 + C$
Square	$\int x^2\, dx$	$x^3/3 + C$
Reciprocal	$\int (1/x)\, dx$	$\ln\lvert x\rvert + C$
	$\int (1/(ax+b))\, dx$	$(\ln\lvert ax+b\rvert)/a + C$
Exponential	$\int e^x\, dx$	$e^x + C$
	$\int e^{ax}\, dx$	$e^{ax}/a + C$
	$\int a^x\, dx$	$a^x/\ln(a) + C$
	$\int \ln(x)\, dx$	$x \ln(x) - x + C$
Trigonometry (x in radians)	$\int \cos(x)\, dx$	$\sin(x) + C$
	$\int \sin(x)\, dx$	$-\cos(x) + C$
	$\int \sec^2(x)\, dx$	$\tan(x) + C$
Rules	**Function**	**Integral**
Multiplication by constant	$\int cf(x)\, dx$	$c\int f(x)\, dx$
Power Rule ($n\neq-1$)	$\int x^n\, dx$	$x^{n+1} / (n+1) + C$
Sum Rule	$\int (f + g)\, dx$	$\int f\, dx + \int g\, dx$
Difference Rule	$\int (f - g)\, dx$	$\int f\, dx - \int g\, dx$